

17th Mar 26 / DA6360

⇒ (Property II contd.)

- even if we don't make a full step, even an incremental update is fine.

↳ because it cancels the curvature.
we can be said equalize the curvature.

Issue $(\nabla^2 f(x))^{-1}$ is costly, since, matrix inversion is required.

↓
Solⁿ

Quasi Newton Methods

$$M_t \approx \nabla^2 f(x_t) \quad M_t^{-1} \text{ should be easy}$$

$$M_t \approx (\nabla^2 f(x_t))^{-1}$$

$$M_t \approx \text{diag}(\nabla^2 f(x)^{-1})$$

⇒ RMS Prop

store a velocity $v_{t+1}(i^*) = \beta v_t(i) + (1-\beta) \frac{[\nabla f(x_t)(i)]^2}{\text{term}}$

$$x_{t+1}(i) = x_t(i) - \alpha \frac{\nabla f(x_t)(i)}{\sqrt{v_{t+1}(i) + \epsilon}}, \quad i = 1, \dots, d$$

$$M_t \approx \begin{bmatrix} \frac{1}{\sqrt{v_1(i)}} & \dots & \frac{1}{\sqrt{v_d(i)}} \end{bmatrix}$$

⇒ Adam (RMS Prop + Momentum)

$$m_{t+1} = \beta_1 m_t + (1 - \beta_1) (\nabla f(x_t)) \quad \text{memory of past gradients}$$

$$v_{t+1}(i) = \beta_2 v_t(i) + (1 - \beta_2) [\nabla f(x_t)(i)]^2$$

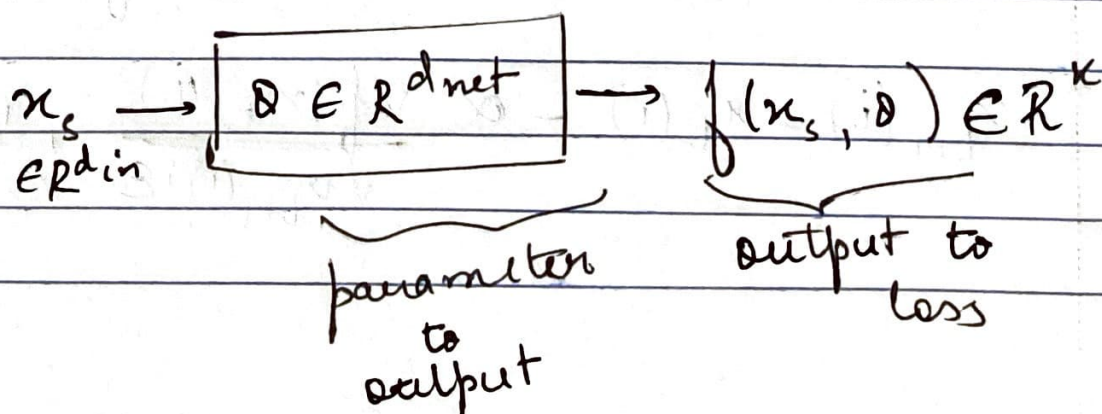
$$m = \frac{m_{t+1}}{1 - \beta_1^{t+1}}$$

$$v(i) = \frac{v_{t+1}(i)}{1 - \beta_2^{t+1}}$$

$$\beta_1, \beta_2 \in [0, 1]$$

$$x_{t+1}(i) = x_t(i) - \alpha \frac{m}{\sqrt{v(i)} + \epsilon}$$

⇒ Gauss Newton Matrix



$$f(x_s, \theta) = \begin{bmatrix} f_1(x_s, \theta) \\ \vdots \\ f_k(x_s, \theta) \end{bmatrix}$$

$$L(\theta) = \frac{1}{S} \sum_{s=1}^S l(f(x_s, \theta), y_i)$$

loss on each eq. input features ground truth

Jacobian

$$J_f = \begin{bmatrix} \frac{\partial f_1(x_s, \theta)}{\partial \theta_1} & \dots & \frac{\partial f_1(x_s, \theta)}{\partial \theta(d_{\text{net}})} \\ \vdots & & \vdots \\ \frac{\partial f_k(x_s, \theta)}{\partial \theta_1} & \dots & \frac{\partial f_k(x_s, \theta)}{\partial \theta(d_{\text{net}})} \end{bmatrix}$$

$\in R^{k \times d_{\text{net}}}$

$$\nabla_{\theta} l(f(x_i, \theta, y_s)) = \nabla_{\theta} l(\theta) = \begin{bmatrix} \frac{\partial l}{\partial \theta_1} \\ \vdots \\ \frac{\partial l}{\partial \theta(d_{\text{net}})} \end{bmatrix}$$

$\in R^{d_{\text{net}}}$

$$\nabla_f l = \begin{bmatrix} \frac{\partial l}{\partial f_1} \\ \vdots \\ \frac{\partial l}{\partial f_k} \end{bmatrix} \in R^k$$

$$\partial_{\theta(i)} l(f(x_s; \theta), y_s) = \sum_{k=1}^K \frac{\partial l}{\partial \theta_k} \frac{\partial f}{\partial \theta(i)}$$

$$= \begin{bmatrix} \frac{\partial f_1}{\partial \theta(i)} & \dots & \frac{\partial f_K}{\partial \theta(i)} \end{bmatrix} \begin{bmatrix} \frac{\partial f_1 l}{\partial \theta(i)} \\ \vdots \\ \frac{\partial f_K l}{\partial \theta(i)} \end{bmatrix}$$

$$= (J_f(i))^T \nabla_f l$$

$$- \partial_{\theta(ij)} \partial_{\theta(i)} l = \partial_{\theta(ij)} \left((J_f(i))^T \nabla_f l \right)$$

$$= \left[\sum_{k=1}^K (\nabla^2 f_k) \cdot \frac{\partial f_k l}{\partial \theta(i)} \right] \underbrace{\left[J_f^T \nabla_f^2 f J_f \right]}_{\text{Gauss Newton Matrix}}$$

Matrix